UNIVERSAL B-H CURVE TRACER

1. INTRODUCTION

Ferromagnetic materials contain large number of small regions, called domains. In each domain all the atomic magnets are locked in rigid parallelism. Thus each domain has a net magnetization in a particular direction. When the specimen is exposed to an external magnetic field **H**, the domains with magnetization component along the direction of the field grow at the expense of those, which are not favorably oriented. Thus, the magnetization increases with field as shown in Fig. 1. When all the domains are aligned in the field direction, the specimen gets magnetically saturated.

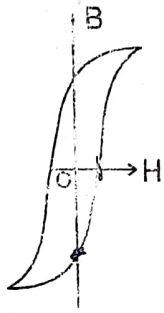


Figure 1 Hysteresis loop

When the external field is removed the domain boundaries do not move completely back to their original positions and as such lead to remanent magnetization (B_r). This in turn leads to a phase lag between the field and the magnetization when the specimen is exposed to an alternating magnetic field. This phase difference between **B** and **H** causes the hysteresis loop. The tendency of the domains to turn around gives rise to mechanical stresses in the specimen; these in turn produce heating. The energy wasted as heat due to cyclic magnetization is given by the area enclosed by the hysteresis loop. This area depends on how far the material is taken into saturation in each cycle and on the properties of the magnetic material. Thus, a study of the hysteresis loop of different magnetic materials helps us to understand their magnetic properties.

From a study of the B-H loops of steel and soft iron we get the following information :

(a) The area of the loop for steel is greater than that for soft iron indicating higher energy loss for steel.

(b) The curve is steep for soft iron. It indicates quick magnetization. Further, the magnetizing field necessary to saturate soft iron is smaller than for steel. Permeability and susceptibility are comparatively large for soft iron.

(c) Steel has greater retentivity.

(d) The coercivity is greater for steel than that for soft iron.

In the conventional techniques of tracing the B-H loop, one has to wind a primary and a secondary coil on the specimen and experiment with it. This method is not convenient for a quick study of the shapes of B-H loops of different materials. The present technique can be used to study B-H loop by simply inserting the specimen in a magnetizing coil. It makes use of a specially designed integrated circuit probe to measure the flux density **B**. The current flowing through the magnetizing coil develops a potential difference across the resistance R which deflects the beam in the X-direction. The deflection is proportional to the magnetic field **H** which is given by :

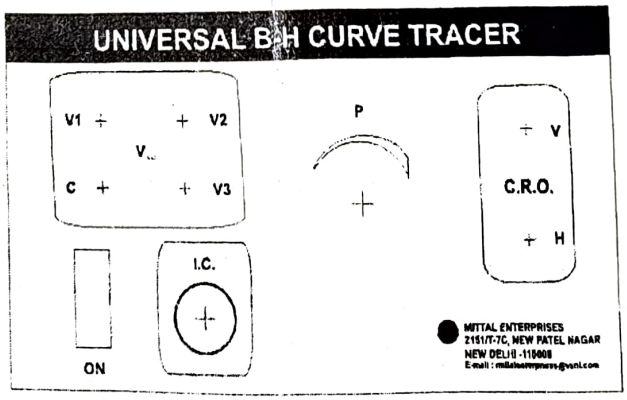
$$H = \frac{N \cdot V_X}{R \cdot L} \text{ ampere turns/meter} \qquad \ldots(1)$$

where N is the no. of turns of the coil, R the resistance in series with the coil i.e. resistance between terminals G & H and L is the coil length in meters. V_X is the voltage applied to the X-input of the CRO. The probe has a sensitivity of 5 mV per Gauss. Hence

$$B = 0.5V_y \text{ weber/m}^2 \qquad \ldots(2)$$

where V_y is the voltage applied to the Y input of the CRO.

2. DETAILS OF THE INSTRUMENT



The instrument consists of:

(a) Main Unit consisting of a variable a.c. supply (marked Va.c.), a resistor R in series with a potentiometer (**P**) and input terminal for the CRO.

(b) Unit housing the I.C. probe with associated circuitry and One magnetizing coil (No. of turns = 300 of SWG26; length of the coil = 0.0323 m).

(c) Samples: 5" nail, ferrite rod and transformer stampings.

3. ADDITIONAL APPARATUS REQUIRED

Any low frequency Laboratory Oscilloscope with horizontal amplification and gain control.

4. EXPERIMENTAL PROCEDURE

(a) Connect one terminal of the magnetising coil to point C of main unit and the other terminal to any of the point between V1 (6 Volt) . Connect **X** to the horizontal input of the C.R.O. and **Y** to vertical input of the C.R.O.

(b) Connect the IC probe to the "IC" marked on main unit .

(c) Switch ON the kit. To get proper loop vary the resistance to the maximum value with the help of knob on the panel.

(d) With no specimen through the coil, adjust the horizontal gain of the C.R.O. until a convenient X deflection is obtained.

(e) Insert a magnetic specimen, e.g. a 5" nail, stampings or ferrite through the magnetising coil such that it touches the probe at the center. **Make sure that sample is touching IC only and conducting tracks should not be shorted in any case.** Adjust the oscilloscope vertical gain (Y gain) until a trace showing the B-H loop conveniently fills the screen.

If the curve is back to front, reverse the connection of the magnetizing coil.

(f) Vary the a.c. voltage by connecting coil between points C and V2 (9 volts) and V3 (12V). The part of the curve showing saturation changes in length.

(g) Trace the area of the loop on butter paper from the screen of C.R.O. and measure the area with the help of graph paper.

(j) The energy loss $= \frac{0.5 \times N}{R \times L} \times S_V \times S_H \times$ area of the loop

Joules/second/Unit volume

where S_V and S_H are vertical and horizontal sensitivities of the C.R.O. for that particular setting of the gains.

(k) Repeat the experiment with different specimens and note your comments on the properties of different materials.

5. PRECAUTIONS

(a) The specimen should touch the probe.

(b) If the area of the loop is expressed in cm^2,the sensitivities should be expressed in volts/cm. If the area of the loop is expressed in sq. meters the sensitivities should be expressed in volts/meter. In either case, the length of the coil should be in meters.

6. WARRANTEE:

This INSTRUMENT is warranteed for one year against any manufacturing defect. **However I.C. probe is not covered under this warrantee**

7. SAMPLE CALCULATION

In a particular experiment

$N = 300$ turns,

$L = 0.033$ meters

a) 5" nail:

$R = 55\ \Omega$

$S_v = 0.5$ V/cm,

$S_h = 2$ V/cm

Area of the loop = 445×10^{-2} cm^2

Energy loss = 367.76 J/Cycle/Unit Volume.

b) Ferrite Rod:

$R = 55\ \Omega$

$S_v = 0.5$ V/cm,

$S_h = 1$ V/cm

Area of the loop = 82×10^{-2} cm^2

Energy loss = 27.78 J/Cycle/Unit Volume.

c) Hacksaw Blade:

$R = 55\ \Omega$

$S_v = 0.1$ V/cm,

$S_h = 2$ V/cm

Area of the loop = 457×10^{-2} cm^2

Energy loss = 75.53 J/Cycle/Unit Volume.

d) Transformer Stamping:

$R = 55\ \Omega$

$S_v = 0.2$ V/cm,

$S_h = 1$ V/cm

Area of the loop = 155×10^{-2} cm^2

Energy loss = 25.62 J/Cycle/Unit Volume.